

EMMANUEL CHIMEZIRIM SAMPSON

Location: Williamsburg, VA 23185, USA **Phone:** +1 (804) 926-0822 **Email:** ecsampson03@gmail.com

LinkedIn: [linkedin.com/in/emmanuel-sampson](https://www.linkedin.com/in/emmanuel-sampson) **GitHub:** github.com/emmanuelcsam **Portfolio:** emmanuelcsam.github.io/portfolio

EDUCATION

William & Mary

Expected May 2026

Bachelor of Science in Physics; Bachelor of Arts in Philosophy
Williamsburg, VA, USA

Concentration: Engineering Physics and Applied Design

Philosophy Honors: Departmental Honors Candidate (Advisor: William D'Alessandro)

Principles for Maximizing Observations within Scientific Practice

GPA: 3.12/4.00

Relevant Coursework:

- **Physics:** Quantum Mechanics I, Electricity & Magnetism I, Classical Mechanics of Particles and Waves I, Modern Physics, Electronics, Scientific Instrumentation Lab, Stellar Astronomy & Cosmology, Practical Computing for Scientists, Scanning Probe Nanoengineering
- **Engineering:** Introduction to Engineering Design, Mechanics of Materials, Capstone EPAD Project
- **Mathematics:** Calculus I & II, Linear Algebra, Multivariable Calculus for Science & Mathematics

Colonial Heights High School

2018 - 2022

Advanced Diploma | GPA: 3.9/4.00
Colonial Heights, VA

RESEARCH INTERESTS

Applied physics and engineering with focus on fiber optics, optoelectronics, and photonics; Materials science including atomistic spin dynamics modeling and semiconductor development; Computer vision and artificial intelligence applications in defect detection and pattern recognition; Advanced instrumentation development for mass spectrometry and electron microscopy; Signal processing for RF engineering and sensor development; Interdisciplinary applications in medical technology and autonomous systems.

RESEARCH EXPERIENCE

Engineer Intern

May 2025 - Present

WEINERT Fiber Optics Inc, Williamsburg, VA

- Designed and implemented AI-powered computer vision systems in Python and C++ for automated fiber-optic end-face defect detection, enhancing transmission efficiency for medical applications including kidney stone removal instrumentation and cancer treatment devices
- Developed, trained, and deployed Convolutional Neural Networks (CNNs) using High-Performance Computing (HPC) clusters to accurately locate and identify microscopic surface defects
- Engineered custom microscope hardware and proprietary software program that automated the full inspection process
- Developed and optimized real-time medical temperature monitoring and logging sensors for surgical applications
- Executed transmission testing and polarization alignment of stress rods for Polarization-Maintaining (PM) Fiber

- Reconfigured and managed industrial robotics, programming user-interface and internal system integration

Physics Research Assistant

October 2023 - Present

Applied Research Center Core Labs, William & Mary, Newport News, VA

- Conducting fundamental research in reflective optics and atomistic spin dynamics modeling for next-generation semiconductor applications
- Collaborating with international research institutions (CERN, NIST, CALTECH) on advanced physics projects including low-angle reflectivity experiments for particle detector development
- Assembled and calibrated TRIFT 4 time-of-flight secondary ion mass spectrometry (TOF-SIMS) instrument for high-resolution surface chemical analysis
- Operating advanced characterization instrumentation: Hitachi S-4700 Field Emission SEM, Bruker Dimension ICON AFM, HIROX KH-7700 3D Digital Microscope, Renishaw inVia Raman Spectrometer, and surface profilers
- Managed \$4,000 budget for procurement of electronics and laboratory equipment
- Designed and conducted reflective-optics experiments measuring infrared light reflectivity at shallow angles
- Training graduate students and industry professionals on operation and theory of material characterization techniques

Student Makerspace Engineer

October 2022 - October 2023

W&M Makerspace, Williamsburg, VA

- Pioneered establishment of 3D Scanning laboratory for digital preservation of anthropological artifacts
- Managed and resolved over 1,000 service requests through Microsoft Jira system
- Trained and certified over 200 students on safety protocols and operation of laser cutters, 3D printers, CNC mills
- Designed and led multi-level robotics workshop series with tiered curriculum
- Maintained fleet of 19 3D printers, 150W CO2 Boss Laser cutter, and 8 sewing machines

PROJECTS

AI Fiber-Optic Defect Analyzer

May 2025 - Present

WEINERT Fiber Optics

Engineered custom microscope and software to automatically locate and identify surface defects on fiber-optic end-faces. Implemented computer vision algorithms in defect classification, enhancing transmission efficiency for medical applications in minimally invasive surgery.

Atomistic Spin Dynamics Modeling

October 2023 - Present

William & Mary

Investigating optimal configurations of iron-nickel and cobalt-iron alloys to enhance magnetization properties as cost-effective alternatives to rare-earth metal semiconductors. Utilizing VAMPIRE software for atomistic simulations of magnetic behavior at nanoscale.

WaveFinderAI - Microplastic Detection System

Hackathon 2025

2nd Place Winner

Designed machine learning tool to detect and classify microplastic contaminants in fluid systems by analyzing unique reflective frequencies. Made to achieve real-time analysis of fluid velocity, contaminant size/shape, and concentration levels with applications in predictive maintenance for industrial hydraulics.

Bird Sensor Technology

William & Mary Hackathon 2024

1st Place Winner

Developed an innovative IoT sensor system detecting bird-window collisions through frequency analysis.

Engineered predictive model using electromagnetic signatures of bird flight patterns with real-time collision avoidance system.

LEADERSHIP EXPERIENCE

President

2023 - 2025

William & Mary Robotics Club

- Designed and delivered semester-long curriculum covering analog/digital electronics, RF devices, and control systems
- Led weekly workshops for 20+ students with tiered instruction from beginner to advanced levels
- Developed hands-on projects including autonomous robots and IoT devices

President

2024 - 2025

William & Mary Data Science Society

- Facilitated weekly technical workshops on Power BI, Kaggle, ArcGIS, and Tableau
- Designed and launched advanced workshops on Computer Vision, Machine Learning, and Neural Networks

Additional Leadership Roles

Executive Positions

- President, Tech Executive Association (2024-2025)
- President, Photo Club (2024-2025)
- Outreach Director, Society of Physics Students (2024-2025)
- Outreach Director, Association of Computing Machinery (2023-2024)

Committee & Service

- Challenge Host, Association of Computing Machinery (2023)
- NASA Volunteer, Langley Research Center (2023)

HONORS, AWARDS & RECOGNITION

Academic Honors

- William & Mary Spirit Scholarship (2024)
- William & Mary Catron Scholarship (2024)
- Jennifer Bosanko Memorial Scholarship (2024)
- GRASP College Success Scholarship (2025)
- Sharpe Community Scholar (2022-2026)
- National Residence Hall Honorary (2025) — *Top 1% of residence hall leaders*
- Philosophy Departmental Honors (2025-2026)

Competition & Recognition

- W&M Hackathon Winner (2024, 2025)
- WaveFinderAI Hackathon — *2nd Place Winner* (2025)

TECHNICAL SKILLS

Programming Languages

Proficient: Python, MATLAB, C++, Java, JavaScript, HTML/CSS, Visual Basic, R, SQL, Arduino/C

Software & Tools

- **Engineering:** Autodesk Suite (Fusion360, AutoCAD), SolidWorks, TinkerCAD, KiCad, Autodesk Eagle, Multisim, Ansys HFSS, COMSOL

- **Data Analysis:** Tableau, Power BI, ArcGIS, SPSS, Kaggle, Jupyter Notebooks
- **Scientific Computing:** High Performance Computing clusters, LabVIEW, VAMPIRE

Laboratory Instrumentation

- **Electron Microscopy:** Hitachi S-4700 FE-SEM, Phenom SEM
- **Scanning Probe Microscopy:** Bruker Dimension ICON AFM
- **Mass Spectrometry:** TRIFT 4 TOF-SIMS
- **Optical Characterization:** HIROX KH-7700 3D Digital Microscope, Renishaw inVia Raman Spectrometer
- **Surface Analysis:** Dektak 3 ST Surface Profiler, Profilm3D

Specialized Technical Skills

Fiber optics characterization • Computer vision & CNNs • Signal processing • RF engineering • PCB design • Circuit analysis • Data acquisition systems • Vacuum systems • Cleanroom protocols • Additive manufacturing (3D printing) • CNC machining • Laser cutting

CERTIFICATIONS

- Introduction to Python Training — The Boehly Center for Excellence in Finance
- From Excel to Tableau — Data Visualization Certification
- Tableau for Data Scientists — Advanced Analytics
- Corporate Finance: Environmental, Social, and Governance (ESG)
- Laser Safety Certification
- Chemical Safety Training

LANGUAGES

- **English:** Native proficiency
- **Igbo:** Native proficiency
- **Japanese:** Conversational (4 semesters formal study)

PRESENTATIONS

Conference Presentations

- “Atomistic Spin Dynamics in Iron-Nickel Alloys” — William & Mary Undergraduate Research Symposium (2026)
- “Building Convolutional Neural Networks For Computer Vision Within High Performance Computing Clusters” — Super Computing (SC25) (2025)

Technical Reports

- “Low Angle Reflectivity Measurements for Material Characteristics” — Applied Research Center Collaboration with CERN, NIST, CALTECH (2024)
- “Optimization of Fiber-Optic End-Face Inspection Using Machine Learning” — WEINERT Internal Report (2025)

REFERENCES

Dr. Douglas Beringer

Head Research Engineer

Applied Research Center, William & Mary

Email: dbberinger@wm.edu

Supervisor for research conducted in the Applied Research Center

Dr. David Lahneman

Production Engineering Manager

WEINERT Fiber Optics Inc.

Email: david.lahneman@weinert-industries.com

Direct supervisor for fiber optics research and AI development

Dr. Dennis Mannos

Vice Provost of Research at William & Mary

William & Mary

Email: dmmanos@wm.edu

Research director of the Applied Research Center